

Speech Production

Speech is a unique, complex, dynamic motor activity through which we express our thoughts and emotions. It consists of a lot of information including **linguistic information (message)** and **non-linguistic information** (speaker identity, gender, language, speaker environment, speaker emotional state and speaker health condition). Speech production is the process by which thoughts are translated into speech.

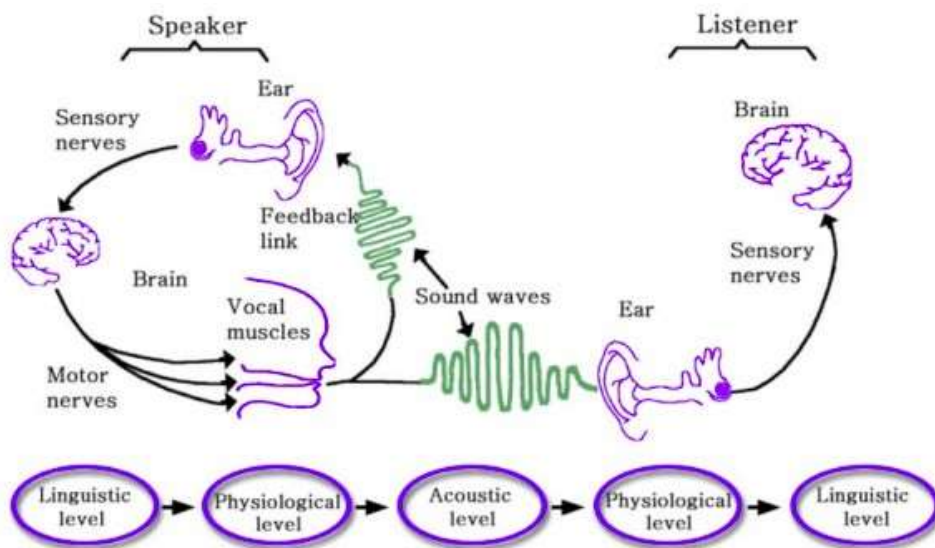


Figure 1. Speech production chain

Figure 1 represents the speech chain, which describes the various levels of processes involved in speech production and speech perception. In this process of speech production, thoughts and emotions are converted to a linguistic message, then corresponding to the message, speech motor programming is done. Then motor execution will be done to produce the speech sound, which corresponds to the message. While speech production, the auditory perception system acts as a feedback mechanism.

Another way of understanding speech production is as, “compression of the lung volume causing air flow, which may be made audible if set into vibration by the activity of the **larynx**. This sound can then be made into speech by various modifications of the supralaryngeal **vocal tract**”. Main activities in speech production can be summarised as,

1. Speech is achieved by Lungs provide the energy source - **Respiration**
2. Vocal folds convert the energy into audible sound - **Phonation**
3. Articulators transform the sound into intelligible speech - **Articulation**

Rhythm, Duration of utterance,

Physiology of speech production:

Speech production requires airflow from the lungs (respiration) to be phonated through the vocal folds of the larynx (phonation) and resonated in the vocal cavities shaped by the jaw, soft palate, lips, tongue and other articulators (articulation)[Wikipedia]. It is convenient to divide the speech production system into three parts:(1) the system below the larynx (**respiration system**), (2) the **larynx** and its surrounding structures, and (3) the structures and the airways above the larynx (**vocal tract**).

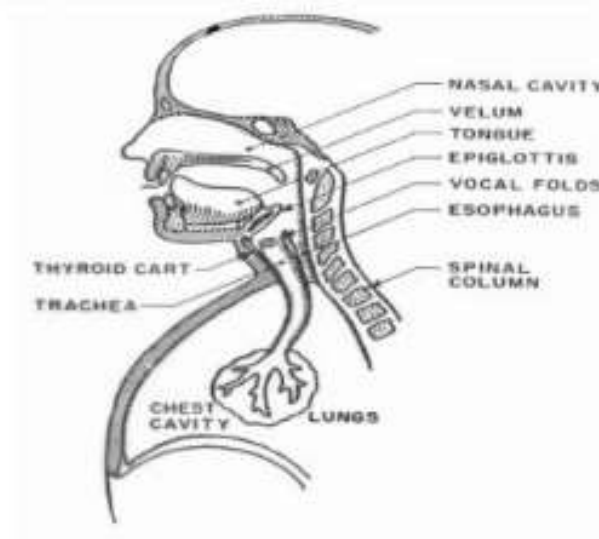


Figure 2. Physiological model of speech production

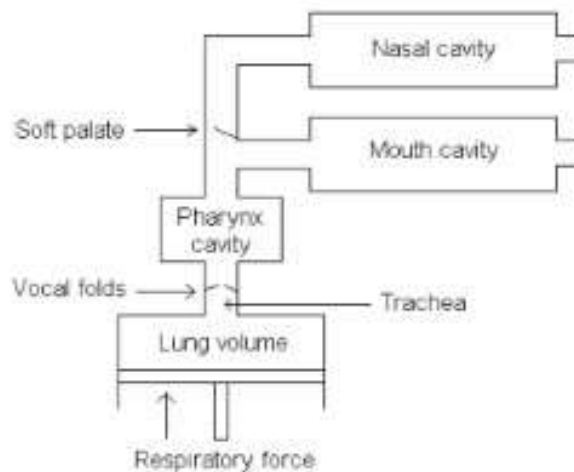


Figure 3. A simplified diagram of the vocal tract showing the energy generation function of the

lungs. Above the vocal folds are the various cavities that can be modified in size and shape to cause changes in the sound quality of the speech sounds.

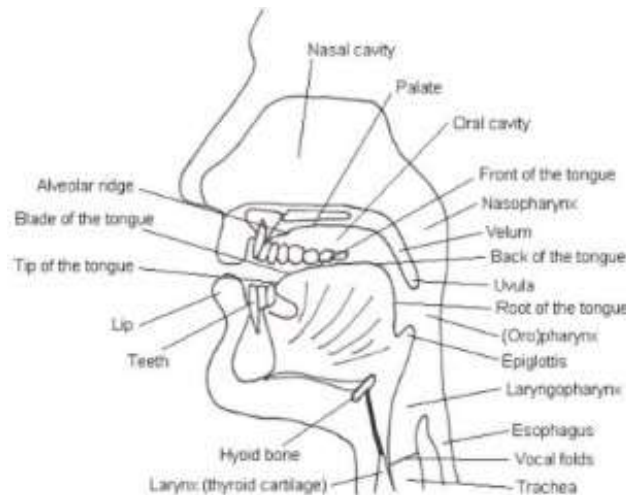


Figure 4. An overview of the vocal tract showing structures that are important in speech sound production and speech articulation.

The respiratory system: Most massive slowly-moving structures provides energy for sound production has fluctuations to help signal emphasis but these variations are at relatively constant level of subglottal pressure. Different patterns of respiration: breathing, reading aloud, spontaneous, counting. Different muscles are active at different phases of the respiratory cycle – a complex, low-level motor program.

Larynx: It is smallest structures, most rapidly contracting muscles. The activity vocal fold vibration is called as voicing. Voicing, turned on and off segment-by-segment. So that that speech has voiced and unvoiced regions in it. Fundamental frequency (F0) is accounted to suprasegmental regulation of vocal fold vibration.

Vocal tract: It is in intermediate-sized, have slowly moving structures like tongue, lips, velum, mandible also called as articulators. One can produce sounds at rates up to 15/sec. The rate of articulation is also correlated to this. Based on the articulators position and how they change the acoustic space of vocal tract, the characteristics of articulation can be named as **place of articulation (POA)** and **manner of articulation (MOA)**.

Note: Each sound unit (**phoneme**) can be characterized by its POA, MOA, Voicing. **Fundamental frequency (F0)** is lowest harmonic present in voiced sound unit. The average F0 value for males is in the range of 100-200 Hz and for females is in the range of 180-300 Hz.

Based on the dimensions of speech production like articulation POA, MOA, voicing speech broadly

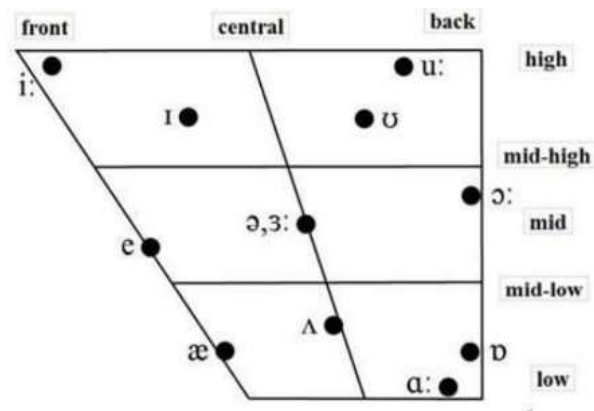
classified into two **vowels** and **consonants**.

Vowels:

A sound that is pronounced with opened mouth, no contact between the tongue and the top of the mouth or teeth and no obstruction to the flow of air. We can use five letters of English represent the vowel sound: a, e, i, o, u. Variations of these sounds makes 20 vowels in total: 7 short vowels, 5 long vowels, 8 diphthongs.

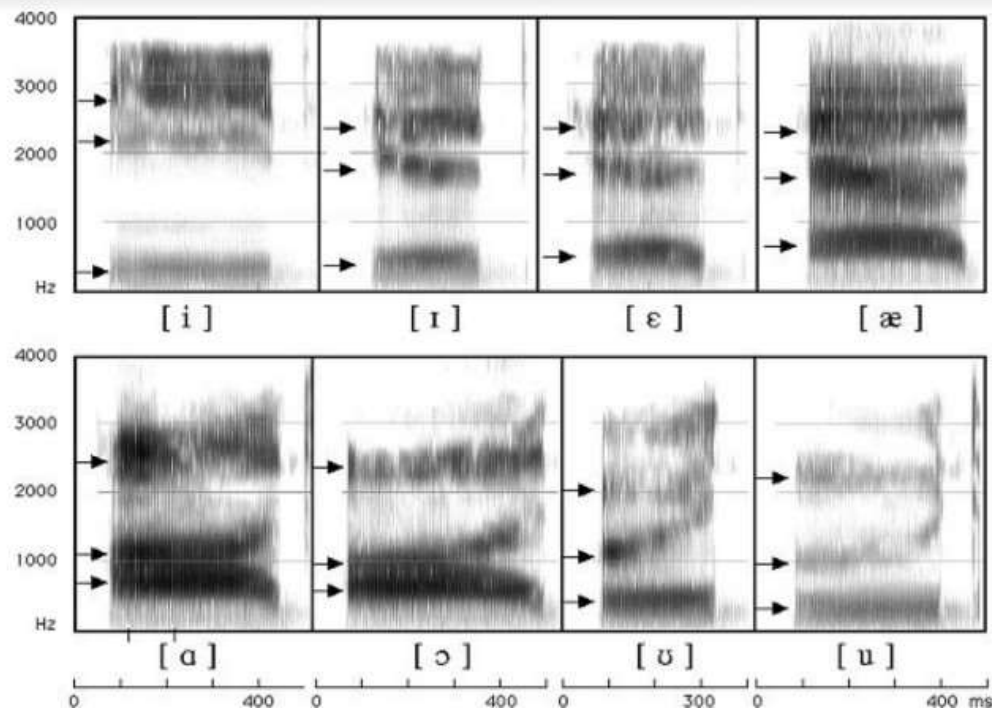
Vowels are divided into different categories based on

- Length of the sound
- Position of the tongue
- Shape of the lips



What is Formant?

A formant is a concentration of acoustic energy around a particular frequency in the speech wave. There are several formants, each at a different frequency, roughly one in each 1000Hz band. Each formant corresponds to a resonance in the vocal tract. We distinguish one vowel from another by the differences in these overtones. According to Lagefoged (2006), each vowel has three formants, i.e. three overtone pitches. The first formant (F1) is inversely related to vowel height. The second formant is related to the degree of backness of a vowel. Formants can be seen in a **wideband spectrogram** as dark bands.



Consonants: Three Dimensions of Consonants

- Ø Place of articulation
- Ø Manner of articulation
- Ø Voicing

Place of Articulation:

- **Bilabial** - use of both lips, e.g. /p/, /b/, /m/ and /w/
- **Labio-dental** - upper teeth and lower lip /f/ and /v/
- **Dental** - upper and lower teeth and tip of tongue /T/ and /D/
- **Alveolar**- tip of tongue touches ridge behind teeth /t/ and /d/, /n/ and /l/; blade of tongue touches ridge behind teeth , /s/ and /z/
- **Palato-alveolar (post-alveolar)** – tongue touches the front part of the palate (roof of mouth) /S/ /Z/ /tS/ /dZ/ /r/
- **Palatal**- tongue touches the middle of the palate /j/
- **Velar** -back part of tongue touches soft palate /k/ and /g/, /N/
- **Glottal**-opening in vocal tract when vocal chords part /h/

Manner of Articulation:

- **Plosive / stop:** Air stream is blocked completely à release of air e.g. /p/, /b/, /t/, /d/, /k/, and /g/
- **Fricative:** Air escapes through a small passage à hissing sound e.g. /f/, /v/, /T/, /D/, /s/, /z/, /S/, /Z/, and /h/
- **Affricate:** plosive + fricative e.g. /tS /and /dZ/
- **Nasal:** Air exits thru the nose e.g. /m/, /n/, and /N/
- **Lateral:** Air exits along the sides of the tongue e.g. /l/

Consonant classification chart based on MOA, POA and Voicing nature:

Place of articulation	Manner of articulation				Nasals	Semi vowels	Fricatives
	Unvoiced		Voiced				
	Unaspirated	Aspirated	Unaspirated	Aspirated			
Velar	k	kh	g	gh	kn		h
Palatal	ch	chh	j	jh	chn	y	sh
Alveolar	T	Th	D	Dh	Tn	r	shh
Dental	t	th	d	dh	n	l	s
Bilabial	p	ph	b	bh	m	v	

Different types of speech: (Will be discussed in class)

Relation between articulatory signatures to Acoustic Signatures:

It is known that each sound has its own POA, MOA and voicing type. These characteristics of sound unit are called as articulatory signatures or **articulatory guestors**. From the acoustic signal point of view, each speech sound has different formant frequencies which are unique for those sound units. These formant frequencies are known as acoustic features or acoustics vector or acoustic signature. All different acoustic vector (Formants of sound unit) together formed an acoustic space. Each sound based on their POA and MOAs each sound unit is mapped to acoustic space.

LTI Model of speech production: (Will be discussed in class)

Self-check:

1. How sounds can be produced? Vibration
2. For speech production what is vibrating? Vocal folds
3. What is Formant? Spectral peak or concentration of acoustic energy
4. What is a Phoneme? Smallest sound unit
5. What is source of speech production? "Air flow from lungs" or "Intermittent air flow by vocal folds".

6. What is role of vocal tract in speech production? Resonances or wave shaping.
7. What is meant by voicing?
8. What is meant by place of articulation?
9. What are all different manner of articulations ?
10. For sound /k/ what are POA, MOA, excitation type or voicing nature?
11. What is coarticulation?
12. What are all the stop sounds?
13. What are all different articulatory signatures? POA and MOAs
14. What are all different acoustic signatures? Formants
15. What is the relation between articulatory signatures to Acoustic Signatures?
16. What are all different types of speech? Based on voicing, Voiced speech and unvoiced speech.
17. What are all different assumptions taken in LTI model of speech? Quasi periodic nature, in 20 msec interval of speech vocal tract configuration is constant.
18. What are advantages in considering LTI model of speech? Vocal tract can be modeled as filter. Speech production can be as source filter model analogy. The pulses of air flow from vocal cords are considered as source and vocal tract shape (Specific Formants) represents the filter
19. What is pitch?
20. What is fundamental frequency?

References:

https://ocw.mit.edu/courses/health-sciences-and-technology/hst-722j-brain-mechanisms-for-hearing-an-lecture-notes/6_mot_con_sp_per.pdf
<https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-345-automatic-speech-recognition-lecture-notes/lecture2.pdf>
https://www.google.com/url?sa=D&q=http://ec-concord.ied.edu.hk/phonetics_and_phonology/wordpress/learning_website/chapter_3_consonants_new.htm&ust=1604066100000000&usg=AOvVaw3UG--NAGe4ydhl=en&source=gmail
<http://educypedia.karadimov.info/library/phon6.pdf>

“A course in phonetics” by Lagefoged

“Acoustic phonetics” by K.N. Stevens

“Digital processing of speech signals” by Rabiner

“Handbook of speech signal processing” Benesty Jacob, Sondhi M. M.